Current approaches to modelling the **spread** of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… **simulation** of the spatiotemporal behaviour of **wildland fire** is … risk evaluation and fire **spread simulation** is discussed. Two … of a `new generation' of fire **spread** models; current trends in …

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Machine learning and deep learning for **wildfire spread prediction**: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… highly performant in predicting fire **behavior**. This paper presents a systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction**, detailing the commonly …

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Trending and emerging prospects of physics-based and ML-based **wildfire spread** models: A comprehensive **review**

H Singh, LM Ang, T Lewis, D Paudyal, M Acuna… - Journal of Forestry …, 2024 - Springer

… aiding in the development of more accurate and reliable **prediction** tools. Methods for literature **review** This **wildfire spread model review** analysed research published in peer-reviewed …

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A **review** of genetic algorithm approaches for **wildfire spread prediction** calibration

J Pereira, J Mendes, JSS Júnior, C Viegas, JR Paulo - Mathematics, 2022 - mdpi.com

… **prediction** based on the use of evolutionary algorithms for input parameter set calibration. In … literature **review**, it was observed that the current literature on **wildfire spread prediction** …

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A systematic **review** and bibliometric analysis of **wildland fire behavior** modeling

J Silva, J Marques, I Gonçalves, R Brito, S Teixeira… - Fluids, 2022 - mdpi.com

… a mathematical percept to **model** the **spread** of **wildland fire**. Bakhshaii and Johnson [7], in a recent **review**, called these last two models a first-generation type of **wildfire** models. The …

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A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire spread** simulation and animation … In this paper, a **review** on **wildfire simulation** models has been presented. Before that, a brief …

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A **review** of **wildland fire spread** modelling, 1990-present 3: Mathematical analogues and **simulation** models

AL Sullivan - arXiv preprint arXiv:0706.4130, 2007 - arxiv.org

… of **review** papers endeavours to critically and comprehensively **review** all types of surface fire **spread** … This paper reviews models of a **simulation** or mathematical analogue nature. Most …

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A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… better anticipate and mitigate the impacts of **wildfire spread**. This **review** will discuss the range of empirical and dynamic tools used for **wildfire prediction**, highlighting their advantages, …

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A **review** of a new generation of **wildfire**–atmosphere modeling

A Bakhshaii, EA Johnson - Canadian Journal of Forest …, 2019 - cdnsciencepub.com

… The earliest physical science models tend to deal with isolated microscale phenomena (eg, **wildfire spread model** or the optical properties of ice crystals) so that their direct application …

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Facing the **wildfire spread** risk challenge: where are we now and where are we going?

J Sun, W Qi, Y Huang, C Xu, W Yang - Fire, 2023 - mdpi.com

… Consequently, it is imperative to comprehensively **review** and explore further … for **wildfire spread simulation**. In Figure 6, various visual elements of **wildfire spread behavior simulation** …

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